Q1 Heuristic design and admissibility

① steps Metric

h(state) = sum (1 for i in range(len(state)) if state[i] != goal[i].

- "Lower" bounds the absolute minimum number of single-letter moves required to reach the goal state

- Every Problem in word ladder Problem replaces at most one char at a time. if the current words is different from the goal in k position, you need at least k separate letter change to fix them. since each cost is 1, taking at least k steps guarantees a cost of at least k. Taking detour words through the dictionary will only add extra moves, so the actual cost will always be ≥ k.
  So, h(state) ≤ h* (state), which make heuristic admissible

Scrabble Metric

- h(state) = sum(Scrabble_Values[goal[i]] for i in range(len(state)) if state[i] != goal[i])

- unavoidable Cost Lower-Bounded:
  It lower bounds the minimum scrabble Point cost you are forced to pay to Place the required goal letters into their correct Positions.

- in scrabble metric, the cost of action is Point value of newly inserted character.
  To Transform current word into goal word, Every mismatched index i must Eventually receive the character goal[i]

Frequency Metric

`h(state) = float(sum(1 for i in range(len(state)) if state[i] != goal[i]))`

Unavoidable cost lower bound:

Lower bounds the remaining path cost assuming every future word visited has the lowest possible rarity score which is rarity = 0

Admissibility justification:

Cost to move to a word is defined as 1 + rarity. The absolute min cost for single move is 1.0 + 0 = 1.

$h(state) \leq h^*(state)$ and heuristic is admissible

②

Word Pair	Cost Metric	UCS Nodes Expanded	A* Nodes Expanded	Node Reduction
Why → Ash	Steps	773	16	97.93%
	Scrabble	765	13	98.30%
	Frequence	808	755	6.56%
Bear Duck	Step	691	6	99.13%
	Scrabble	2981	6	99.80%
	Frequency	1094	720	33.58%

$$\text{Reduction} = \frac{UCS - A^*}{UCS} \times 100\%$$

Mini part

① Yes, UCS and A* always return exact same optimal cost for every word pair across all three cost measure.

② Scrabble produce the largest percentage

③ Frequency was the least formative cost measure.

④ A* search are prioritize using $f(n)=g(n)+h(n)$.

Step and scrabble: both heuristics provided strong, realistic estimates. Scrabble adds up the point values of the goal's missing letters, which represents a huge part of the actual remaining cost. Because $h(n)$ is accurate and significant A* quickly prunes bad paths,

For frequency: The real cost of a move in frequency includes word rarity penalties, often making single moves cost much more than 1.0 However $h_{frequency}$ assumes every moves cost the minimum of 1.0 Because this heavily underestimates the real remaining cost, $h(n)$ is tiny compared to $g(n)$ causing A* to ignore the heuristic and search almost identically to UCS.

you need at least

, taking at least
ds through the
always be ≥ k.

] b=goal[i]]

Q7 ① Step Metric:

Equal in both direction → Yes!,

Every valid transition between two cost is Exactly 1.
If there is a valid path from w_1 to w_2 taking k steps, reversing exact moves a valid path from w_2 to w_1 with the exact same k step.

Counter Example: N/A

② Scrabble Metric!

Equal in both direction → NO!

Transition cost depend on the scrabble point value of the newly inserted letter in destination word. Moving A→B incurs the cost of the letter added to B, while moving B→A incurs the cost of the letter added to A. When swapping two letters with different scrabble values, the edge costs are asymmetric

Counter Ex:

cat → hat: Replace c with h. cost = scrabble value of h = 4

hat → cat: Replace h with c. cost = scrabble value of c = 3

therefore, $c^*(cat, hat) = 4 \neq 3 = c^*(hat, cat)$

③ Frequency Metric:

Equal in both direction → NO

Transition cost is defined as 1 + rarity($w_{destination}$). Moving $w_1 \rightarrow w_2$ pays the rarity penalty of w_2 whereas moving $w_2 \rightarrow w_1$ pays the rarity penalty of w_1. If w_1 and w_2 have different word frequencies in dictionary, transition cost are asymmetric.

Counter Ex: cat → vat: cost = 1 + rarity(vat)
vat → cat: cost = 1 + rarity(cat)

Vat is rarer than cat, rarity(vat) > rarity(cat) making $c^*(cat, vat) \neq c^*(vat, cat)$

Q4

① H(n) in the doc file

Part 5:

Problon ($w_1 \to w_2$)	Heuristic	Solution cost	Nodes Expanded
why → ask	Baseline(h_0)	38.8554	755
why → ask	Repaired($h_{repaired}$)	38.8554	634
bear → duck	Baseline(h_0)	38.2402	720
bear → duck	Repaired($h_{repaired}$	38.2402	702

① Solution cost, Both h produce identical 38.8554 and for why 38.2402

Nodes Expanded:

$h_{repaired}$ reduce expansion from 755 to 634 for why ask

720 to 702 for bear → duck.

Conclusion: Yes this evidence support using $h_{repaired}$ because

$h_{repaired} \geq h_0(n)$ while being admissible.